

Table 1

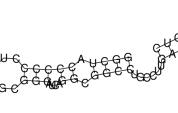
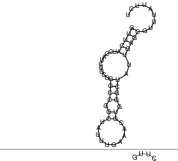
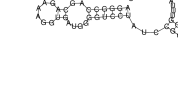
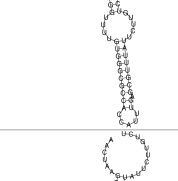
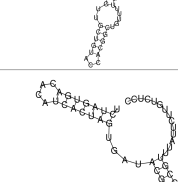
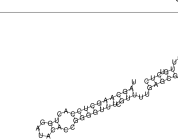
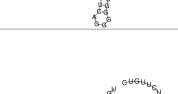
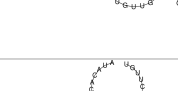
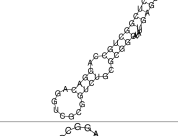
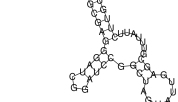
| Name       | Read Count<br>in the sample | RPM (reads per<br>million) | Sequence                                                                | Dot bracket                                                                         | MFE    | GC     | SIM  |
|------------|-----------------------------|----------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------|------|
| Aptamer_0  | 9                           | 49.8091                    | GGCUACCCCCCU<br>UCGCGGGAUCAG<br>GCGGCCUGCCUU<br>GAGCGUUUAUUC<br>UUGUCUC | ((((.<br>((((.....))....)).....<br>((((.....)).))                                   | -11.8  | 61.82% | 1    |
| Aptamer_1  | 9                           | 49.8091                    | AGCUCGUUGCUA<br>AUAAGACAAUGC<br>GCUUUGAGCGUU<br>UAUUCUUGUCUC<br>CC      | ((.....)).((((....<br>((((.....))....)).....<br>((((.....)).))                      | -11.50 | 44.00% | 0.49 |
| Aptamer_2  | 8                           | 44.2747                    | GUUCAUCCAUCU<br>GCGGGCCGGCUA<br>UUUGAAACGUGG<br>GCUAUUGAGCGU<br>UUAUUCU | (((((.....((((.<br>((.....)).))....)).....<br>..                                    | -11.20 | 49.09% | 0.38 |
| Aptamer_3  | 7                           | 38.7404                    | AAGAUCUCGUGU<br>AGGUGGCCGCAU<br>CGCCGGCUCGCC<br>AUUUGAGCGUUU<br>AUUCUUG | .(((((((.(.<br>(((((((.....))....))....))....))<br>)).....                          | -16.60 | 54.55% | 0.45 |
| Aptamer_4  | 6                           | 33.206                     | AGGGCCAGCAGA<br>AAGGUGAUGGGG<br>UCCUAUCCGCUU<br>GAGCGUUUAUUC<br>UUGUCUC | (((((....((.....))....))....))...<br>(((.....)).....                                | -11.30 | 52.73% | 0.58 |
| Aptamer_5  | 6                           | 33.206                     | UAUUUGAGGUUU<br>UGUGGGCGCCAC<br>CAUUUGAGCGUU<br>UAUUCUUGUCUC<br>CC      | .....((((....<br>((((((((.....)).))....))....))<br>)).                              | -10.80 | 46.00% | 0.44 |
| Aptamer_6  | 6                           | 33.206                     | AACUAAGGGCGC<br>GUCGUUGCUGUA<br>CCACGGCUGUUU<br>UGAGCGUUUAUU<br>CUUGUCU | .....(((((((.(.<br>((((.....))....))....))....))<br>....                            | -11.70 | 49.09  | 0.58 |
| Aptamer_7  | 6                           | 33.206                     | UCAGGCCUGGAU<br>GUUGUGGAAAGA<br>UACGAGUAGUAU<br>UGAGCGUUUAUU<br>CUUGUCU | ..((((....((((.(.<br>((((.....))....)).))....))....))<br>((((.....))....)).))....)) | -10.40 | 41.82% | 0.55 |
| Aptamer_8  | 6                           | 33.206                     | UCUAGUGACACA<br>UCACUAGUGAUA<br>CGCCGUUCCUUG<br>AGCGUUUAUUCU<br>UGUCUCC | .(((((((.....))....))....))....<br>((((.....)).)).....                              | -11.60 | 45.45% | 0.62 |
| Aptamer_9  | 6                           | 33.206                     | UAGCAAGCUCCA<br>CUGGAUACACCG<br>GGGUUUCGUUUU<br>GAGCGUUUAUUC<br>UUGUCUC | .((((((((....<br>((.....)).))....))....))....<br>((((.....)).))                     | -10.10 | 47.27% | 0.60 |
| Aptamer_10 | 6                           | 33.206                     | AUUCGACCUGUU<br>CCGGUCUUAUCU<br>UCAUGCUCAAUU<br>GAGCGUUUAUUC<br>UUGUCUC | ....((((.....))....)).....<br>((((.....))....)).....                                | -12.00 | 41.82% | 0.58 |
| Aptamer_11 | 6                           | 33.206                     | AACUUCAACGCA<br>GGUUCAUGGGUC<br>AGGGGGCCUGGG<br>GUUUGAGCGUUU<br>AUUCUUG | (((((((((.<br>(((((((.....))....)).))....))<br>.)).....                             | -12.80 | 52.73  | 0.53 |
| Aptamer_12 | 6                           | 33.206                     | UGGGGGUCGGGU<br>GUGUACCCGGCU<br>AUUGUUGCACGG<br>UUGAGCGUUUAU<br>UCUUGUC | ....(((((((.....))....))....))....<br>(((.....)).....                               | -14.10 | 54.55  | 0.47 |
| Aptamer_13 | 6                           | 33.206                     | AUACACAAACAAC<br>GUUAUCACCGCC<br>GGGGGACGGCCU<br>UUGAGCGUUUAU<br>UCUUGU | .....(((((((.....<br>((((.....))....))....))....<br>((((.....))....))....))         | -11.50 | 49.09  | 0.47 |

| Name       | Read Count<br>in the sample | RPM (reads per<br>million) | Sequence                                                                 | Dot bracket                                                   | MFE    | GC    | SIM  |
|------------|-----------------------------|----------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------|--------|-------|------|
| Aptamer_14 | 6                           | 33.206                     | ACGUUACGAGAA<br>UGGACCGUCCUU<br>GAUGAACGAAUU<br>GAGCGUUUAUUC<br>UUGUCUC  | .....((((((((((....<br>((((((.....)))))).))))))))...<br>.     | -14.20 | 43.64 | 0.55 |
| Aptamer_15 | 6                           | 33.206                     | CGCUCGGCUGCC<br>AGGACAGGUCGC<br>GGUCUGCGCGGC<br>AUAUUGAGCGUU<br>UAUUCUU  | (((((((((((.(((.<br>(...).))))).))))).))....<br>....          | -19.80 | 60    | 0.42 |
| Aptamer_16 | 6                           | 33.206                     | AGGCGAUCGGGC<br>GAGGGAUCGGAU<br>CCGGCUAGUAUU<br>GAGCGUUUAUUC<br>UUGUCUC  | .....((((((((((...)).<br>(((.....)).....)))))).               | -10.00 | 54.55 | 0.62 |
| Aptamer_17 | 5                           | 27.6717                    | UUAUUUGAGUUA<br>GU CGCGGGCGCG<br>GGGCCCUUUGAG<br>CGUUUAUUCUUG<br>UCUCCC  | .....(((....<br>((((((.....)))).....))....))..<br>.....       | -11.90 | 53.70 | 0.55 |
| Aptamer_18 | 5                           | 27.6717                    | GAAAGUUGUCAC<br>UGUGAUUCCUAA<br>GGGCCGUGGCAA<br>GGCGCCCUUGAG<br>CGUUUAU  | .....(((((.<br>((.....)).))))<br>((((((.....))))).<br>CGUUUAU | -12.90 | 52.73 | 0.35 |
| Aptamer_19 | 5                           | 27.6717                    | AUAUUUUUGUCC<br>AUUGUCUCUGCU<br>CAUGUAUGGGAU<br>AUUGAGCGUUUA<br>UUCUUGU  | .....((.(.<br>((((((((....)).))))).))....<br>.....            | -10.40 | 34.55 | 0.42 |
| Aptamer_20 | 5                           | 27.6717                    | CUCCGAGGGAAC<br>CCGUUAUCUCCC<br>GCAUCGUUGCGU<br>UACACCCUUGAG<br>CGUUUAU  | ...((((((.....<br>((((....))....))))).<br>CGUUUAU             | -10.20 | 54.55 | 0.33 |
| Aptamer_21 | 5                           | 27.6717                    | ACGGCUUGGCUU<br>AGAUGAUUAGGU<br>GCGCCCUCAUCG<br>UUGAGCGUUUAU<br>UCUUGUC  | ...(((((((....<br>(((....).)))))))).<br>.                     | -11.7  | 49.09 | 0.53 |
| Aptamer_22 | 6                           | 33.206                     | AGGUCAUUGCAG<br>GGGCCGUGCUAG<br>AGCUAGGAGAGC<br>GUUUUAUUCUUGU<br>CUCC    | ((.(.(((....)).)).).).<br>((((((.....))))))                   | -11.00 | 53.85 | 0.45 |
| Aptamer_23 | 5                           | 27.6717                    | UGUGGAUAGGGU<br>AGGGAUUAUCCU<br>GGGCCAU AUGCG<br>AGCGUUUAUUCU<br>UGUCUCC | ((((((((((((.....)))))).)).).).<br>((((.....))))....          | -13.70 | 49.09 | 0.51 |

| Name       | Secondary structure |         |
|------------|---------------------|---------|
| Aptamer_0  |                     |         |
| Aptamer_1  |                     |         |
| Aptamer_2  |                     |         |
| Aptamer_3  |                     |         |
| Aptamer_4  |                     |         |
| Aptamer_5  |                     |         |
| Aptamer_6  |                     |         |
| Aptamer_7  |                     | Exclude |
| Aptamer_8  |                     |         |
| Aptamer_9  |                     |         |
| Aptamer_10 |                     | Exclude |
| Aptamer_11 |                     |         |
| Aptamer_12 |                     |         |
| Aptamer_13 |                     |         |

| Name       | Secondary structure |         |
|------------|---------------------|---------|
| Aptamer_14 |                     | Exclude |
| Aptamer_15 |                     |         |
| Aptamer_16 |                     |         |
| Aptamer_17 |                     |         |
| Aptamer_18 |                     |         |
| Aptamer_19 |                     | Exclude |
| Aptamer_20 |                     |         |
| Aptamer_21 |                     |         |
| Aptamer_22 |                     |         |
| Aptamer_23 |                     |         |

## Final Aptamer Candidated targeting HPV-16 - L1

| Name  | Read Count in the sample | RPM (reads per million) | Sequence                                                                 | Dot bracket                                             | MFE    | GC     | SIM  | Secondary structure                                                                   |
|-------|--------------------------|-------------------------|--------------------------------------------------------------------------|---------------------------------------------------------|--------|--------|------|---------------------------------------------------------------------------------------|
| AHL1  | 9                        | 49.8091                 | GGCUACCCCCCU<br>UCGCGGGGAUCAG<br>GCGGCCUGCCUU<br>GAGCGUUUAUUC<br>UUGUCUC | (((<br>(((.....)).....)).....<br>(((.....).)))          | -11.8  | 61.82% | 1    |    |
| AHL2  | 8                        | 44.2747                 | GUUCAUCCAUCU<br>GCGGGCCGGCUA<br>UUUGAAACGUGG<br>GCUAUUGAGCGU<br>UUAUUCU  | (((.....(((<br>((.....).)))..))))....<br>....           | -11.20 | 49.09% | 0.38 |    |
| AHL3  | 7                        | 38.7404                 | AAGAUCUCGUGU<br>AGGUGGCCGCAU<br>CGCCGGCUCGCC<br>AUUUGAGCGUUU<br>AUUCUUG  | ..(((((((<br>(((((((.....))..))))..))<br>)))))).....    | -16.60 | 54.55% | 0.45 |    |
| AHL4  | 6                        | 33.206                  | AGGGCCAGCAGA<br>AAGGUGAUGGGG<br>UCCUAUCCGCUU<br>GAGCGUUUAUUC<br>UUGUCUC  | (((.....(<br>((.....)).....))....<br>(((.....)).....    | -11.30 | 52.73% | 0.58 |    |
| AHL5  | 6                        | 33.206                  | UAUUUGAGGUUU<br>UGUGGGGCCGCAC<br>CAUUUGAGCGUU<br>UAUUCUUGUCUC<br>CC      | .....(((.....<br>((((((((.....).))))))....<br>..))))..  | -10.80 | 46.00% | 0.44 |   |
| AHL6  | 6                        | 33.206                  | AACUAAGGGCGC<br>GUCGUUGCUGUA<br>CCACGGCUGUUU<br>UGAGCGUUUAUU<br>CUUGUCU  | .....(((((((<br>((((.....)).....))))))....<br>.....     | -11.70 | 49.09  | 0.58 |  |
| AHL7  | 6                        | 33.206                  | UCUAGUGACACA<br>UCACUAGUGAUA<br>CGCCGUUCCUUG<br>AGCGUUUAUUCU<br>UGUCUCC  | ..(((((((.....))....<br>(((((((.....).))).....<br>.     | -11.60 | 45.45% | 0.62 |  |
| AHL8  | 6                        | 33.206                  | UAGCAAGCUCCA<br>CUGGAUACACCG<br>GGGUUUCGUUUU<br>GAGCGUUUAUUC<br>UUGUCUC  | ..(((((((((((<br>((.....).))))))..))..<br>(((.....).))) | -10.10 | 47.27% | 0.60 |  |
| AHL9  | 6                        | 33.206                  | AACUUCAACGCA<br>GGUUCAUGGGUC<br>AGGGGGCCUGGG<br>GUUUGAGCGUUU<br>AUUCUUG  | (((((((((((<br>((((((((.....)).....).))<br>)))..))..... | -12.80 | 52.73  | 0.53 |  |
| AHL10 | 6                        | 33.206                  | UGGGGGUCGGGU<br>GUGUACCGGCU<br>AUUGUUGCACGG<br>UUGAGCGUUUAU<br>UCUUGUC   | ....<br>((((((((.....)).....<br>(((.....).)).....       | -14.10 | 54.55  | 0.47 |  |
| AHL11 | 6                        | 33.206                  | AUACACAAACAA<br>CGUUAUCACCGC<br>CGGGGGACGGCC<br>UUUGAGCGUUUA<br>UUCUUGU  | .....((((.....<br>((((.....)).....)).....<br>..         | -11.50 | 49.09  | 0.47 |  |
| AHL12 | 6                        | 33.206                  | CGCUCGGCUGCC<br>AGGACAGGUCGC<br>GGUCUGCGCGGC<br>AUAUUGAGCGUU<br>UAUUCUU  | (((((((((((.....(((<br>((.....).)))..))))....<br>.....  | -19.80 | 60     | 0.42 |  |
| AHL13 | 6                        | 33.206                  | AGGCGAUCGGGC<br>GAGGGAUCGGAU<br>CCGGCUAGUAUU<br>GAGCGUUUAUUC<br>UUGUCUC  | .....(((((((((((.....<br>(((.....)).....))))..          | -10.00 | 54.55  | 0.62 |  |